S3Net: Spectral–Spatial Siamese Network for Few-Shot Hyperspectral Image Classification

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***Abstract*— Deep learning (DL) has shown great potential for hyperspectral image (HSI) classification due to its powerful ability of nonlinear modeling and end-to-end optimization. However, DL models easily get trapped into overfitting due to limited training labels since the labeling process is time-consuming and laborious in a real classification scenario. To overcome this issue, we propose a novel spectral–spatial Siamese network (S3Net) for few-shot HSI classification. First, a lightweight spectral–spatial network (SSN) composed of 1-D and 2-D convolution is proposed to extract spectral–spatial features. Second, S3Net is constructed by two SSNs in dual branches, which can augment the training set by feeding sample pairs into each branch and thus enhancing the model separability. To provide more features for the model, differentiated patches are fed into each branch, where negative samples are randomly selected to avoid redundancy. Finally, a weighted contrastive loss is designed to promote the model to fit in the right direction by focusing on sample pairs that are hardly to be identified. Moreover, another adaptive cross-entropy loss is conceived to learn the fusion ratio of the two branches. Experiments based on three commonly used HSI datasets demonstrate that S3Net outperforms traditional and state-of-the-art DL-based HSI classification methods under a few-shot training scenario. In addition, the weighted contrastive loss and the adaptive cross-entropy loss jointly improve the discrimination power of the model.**

***Index Terms*— Deep learning (DL), few-shot learning (FSL), hyperspectral image (HSI) classification, Siamese network.**

I. Introduction

HYPERSPECTRAL image (HSI) has hundreds of continuous bands with a very high spectral resolution, which can jointly provide rich spectral and spatial information to accurately discriminate similar land-cover types [1]. Therefore, HSI classification has been widely used for precision agriculture, forestry monitoring, mineral survey, and environmental assessment.

Manuscript received January 10, 2022; revised April 12, 2022 and May 14, 2022; accepted June 1, 2022. Date of publication June 8, 2022; date of current version June 21, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41971279 and in part by the Fundamental Research Funds for the Central Universities under Grant B200202012. *(Corresponding author: Zhaohui Xue.)*

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Digital Object Identifier 10.1109/TGRS.2022.3181501

In the past decades, HSI classification has been greatly developed by machine learning and pattern recognition models, including support vector machine (SVM), random forest (RF), multinomial logistic regression, representation learning, and multiple kernel learning [2]–[6]. However, their further explorations may be blocked due to the limitation of low-level handcrafted features.

Deep learning (DL) has been widely used for HSI classification due to the powerful ability of data-driven high-level feature learning [7]–[11]. Some typical forms of DL frameworks are stacked autoencoder (SAE) [12], [13], convolutional neural network (CNN) [14], [15], deep belief network (DBN) [16], recurrent neural network (RNN) [17], and long short-term memory (LSTM) [18], among which CNN has become a critical framework for image classification tasks.

Recently, CNN has been well developed for HSI classification, and various methods can be roughly divided into 1-D CNN, 2-D CNN, 3-D CNN, and some hybrid methods. As a naive method, 1-D CNN processes vector information, which performs 1-D convolution kernel along the spectral dimension to extract spectral feature for classification [19]. However, 1-D CNN ignores the spatial distribution. Later, 2-D CNN is developed, which treats HSI as an ordinary image to extract spatial information, and it usually performs better due to covering the spatial differences between pixels [20], whereas the rich spectral information is not fully utilized in 2-D CNN. To overcome this issue, 3-D CNN is more advocated since it jointly captures the spectral and spatial information using 3-D convolution kernels [21], [22].

Although DL has achieved great success for HSI classification, DL models easily get trapped into overfitting due to insufficient training samples. However, the collection of labeled samples is time-consuming and laborious. Therefore, DL-driven HSI classification faces few-shot learning (FSL) issue [23], i.e., only using few labeled samples for classification [24]. At present, three types of networks, including prototypical network [25]–[28], relation network [29]–[31], and Siamese network [32]–[36], have been developed to solve the FSL issue for HSI classification. In addition, there also have some other related studies focused on FSL for HSI domain adaptation [37]–[39].

Prototypical network is designed to obtain the prototype representations of each class by calculating the mean vector of the support points in metric space and then to perform classification by measuring the distance between prototype and query points. As a seminal work, a spatial–spectral

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prototypical network was proposed to better combine the spatial and spectral information in [25]. Later, in order to transfer samples from the original data space to an embedding-feature space, a global prototypical network was proposed in [26]. However, the quality of labeled samples may affect the feature extraction ability of prototypical network, and thus, an active-learning-based prototypical network was proposed, which combines semisupervised clustering and active learning methods to select and request labels from valuable samples actively [27]. Considering the overfitting issue in the presence of few labeled samples and complex spectral feature distributions, a stable prototypical network was proposed by incorporating dropout regularizer [28].

Relation network consists of two modules: the embedding module used for extracting sample-pair features and the relation module used for calculating relation score between a pair of samples. As one of the earliest attempts to employ the relation network for HSI classification, Rao *et al.* [29] proposed a spatial–spectral relation network. Later, Gao *et al.* [30] exploited relation learning by comparing the similarity between different samples, i.e., the relation score between samples belonging to the same class is high, whereas the relation score between samples belonging to different classes is low. Similarly, a task-specific learning network was proposed for HSI classification by using meta-learning and relation learning to learn unbiased features [31].

Siamese network is composed of two subnetworks with shared weights and the feature similarity is calculated to determine whether, in the input, a pair of samples are matched or not. Currently, the Siamese network has been preliminarily exploited for HSI classification, and our previous work has exploited the triplet network for remote sensing image localization [40]. In [32], a Siamese network composed of 3-D CNN was proposed, which combined contrast information with label information. Similarly, Rao *et al.* [33] proposed a 3-D adaptive spatial–spectral pyramid pooling to enhance the feature extraction ability of the subnetwork. Zhang *et al.* [34] designed a new quadruplet loss function in order to learn a feature space where the distances between samples from the same class are shortened, while those from a different class are enlarged. To further enhance the spectral–spatial feature extraction capability of the subnetwork, Huang and Chen [35] used a two-way Siamese network to fuse spectral–spatial features separately and adopted an adversarial training strategy in the case of limited labeled samples. To make full use of the Siamese network's ability in judging the similarity of samples, Jia *et al.* [36] combined the Siamese network and autoencoder to propose a semisupervised Siamese network. Although many studies have explored the Siamese network for HSI classification, further explorations are still needed to improve the classification performance with few labeled samples.

The core reason for low accuracy of FSL lies in unreliable minimizing experience risk. It is very hard to fully learn the feature distribution of the entire training set from few labeled samples, thus resulting in the distribution error when predicting some unknown data [41]. To solve this issue, we focus on the Siamese network to: 1) design a lightweight Siamese network to reduce parameters, which can weaken the overfitting issue; 2) do data augmentation to learn more information from the samples; and 3) design more powerful loss function to improve the discrimination ability of the network.

In this article, a new lightweight spatial–spectral network (SSN) is designed. First, 1-D convolution and 2-D convolution are used to extract the spectral–spatial information of HSI, and then, global average pooling (GAP) is used to classify for reducing network parameters and model complexity further. Second, spectral–spatial Siamese network (S3Net) is constructed by two branches of SSNs. S3Net amplifies samples by pairing and enhances samples separability through similarity measures. In addition, differentiated inputs with two different sizes are designed for providing more features to the model, and random selection (RS) is used to balance the positive and negative sample pairs in a training process. Finally, a weighted contrastive loss function is proposed to introduce more samples that are hard to identify in training and an adaptive loss function is proposed to learn the ratio of two branches for better fusion purpose.

In summary, the main innovative contributions of this article can be summarized as follows.

1) SSN is proposed to efficiently extract the spectral–spatial features of HSI with small parameter size. The network uses 1-D and 2-D convolutions that reduce by 2/3 of the number of parameters in comparison to 3-D convolution. In addition, GAP is used to replace fully connected layer, and it can also reduce parameters and model complexity for weakening the overfitting issue.
2) S3Net is proposed for small samples problem. In S3Net, two different size inputs are fed into the model and they can provide more features to the model for improving recognition ability. To augment training samples, sample pairing and RS strategies are used to increase and balance the positive and negative samples.
3) A weighted contrastive loss and an adaptive loss are proposed to enhance the classification ability of the S3Net. The former improves the coveraging ability of the Siamese network by increasing hard samples proportion in training, whereas the latter can balance the information of the two branches.

II. Related Work

Siamese network is a dual-branch structure, which can realize data augmentation through sample pairing and learn the difference between samples since the input of Siamese network is a pair of samples (i.e., sample pair). Traditional Siamese network is used to determine whether two samples are matched by two subnetworks with shared weight, which cannot distinguish which class of the sample pair belongs to. In the training process, if the sample pair is matched, the model will reduce the distance between their features. When the sample pair is unmatched, the model will increase the distance. In the testing process, the test sample verifies whether the sample matches by calculating feature similarity. As shown in Fig. 1(a), each sample pair contains two samples, and Feature 1 is extracted by model A, while Feature 2 is

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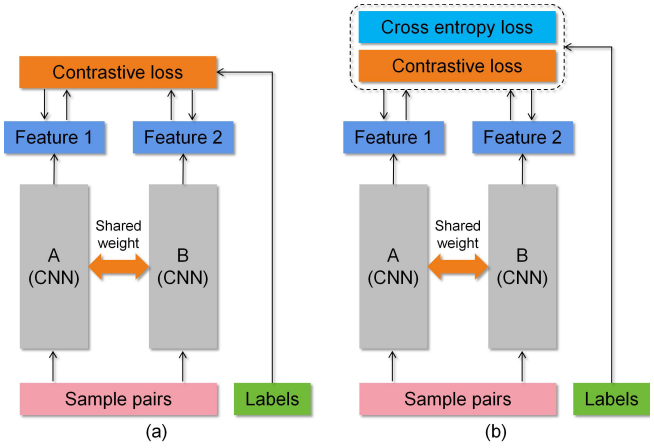


Fig. 1. Structure of Siamese network. (a) Traditional Siamese network. (b) Siamese network for classification. Traditional Siamese network can only distinguish whether two samples are belonging to the sample class, whereas Siamese network for classification can recognize the category of a sample by adding an additional classification structure.

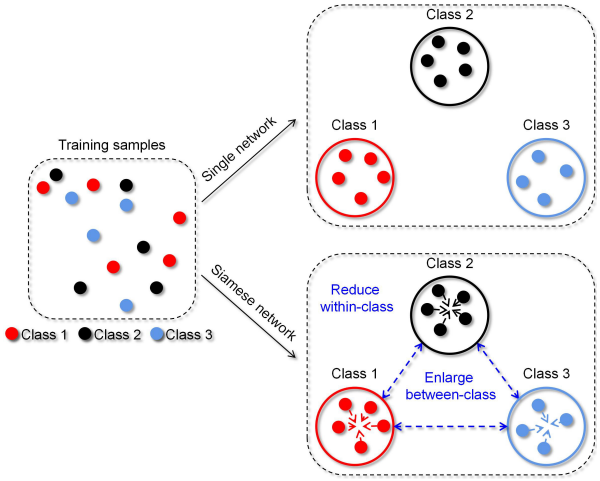


Fig. 2. Comparison of sample relationship between single network and Siamese network.

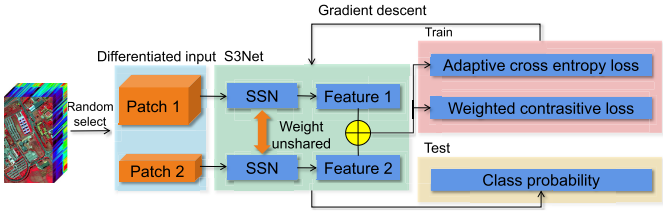


Fig. 3. Training and test mechanisms of S3Net. The input paired samples have different sizes, and we add an unshared weight layer in SSN to adapt different input sizes.

extracted by model B. Contrastive loss (L_{ctr}) is calculated to train the model based on the two features, as shown in the following equation:

$$L_{\text{ctr}}\left(y, d_{\text{pos}}, d_{\text{neg}}\right) = y \times d_{\text{pos}} + (1-y) \times \max\left(m - d_{\text{neg}}, 0\right) \quad (1)$$

where y indicates whether a pair of samples belong to the same class. If they are in the same class, $y = 1$; otherwise, $y = 0$. The distance between sample pair belonging to the same class is denoted by d_{pos} and the distance between sample pair belonging to different classes is denoted by d_{neg}. The margin parameter m is a threshold to prevent negative samples from deteriorating the model when d_{neg} is too large.

Traditional Siamese network belongs to metric model, which may have no classification ability. Two adaptations can be made to implement the Siamese network for classification. The first method is using a labeling strategy. During the training stage, if sample pairs belong to the same class, the sample pair label is the same as the single sample. When sample pairs are from different classes, their label is set to 0. During the testing stage, test samples can be predicted directly based on the output feature vector [33], [35]. The second method is adding a classification structure to the Siamese network, which calculates a cross-entropy loss for each single sample and trains the network for classifying. In the testing stage, a single test sample can be classified directly by the classification structure [32]. As shown in Fig. 1(b), we use the second method to achieve classification, which is more robust since it produces the class probability. In addition, the Siamese network with classification structure can fuse classification and metric learning well to better improve the discrimination power of the model.

Single network only classifies a single sample, which lacks the connection between samples, as shown in the right-top box of Fig. 2. On the contrary, the Siamese network can associate different samples by pairing, which can reduce the distance of within-class samples and enlarge the distance of between-class samples, thereby improving the classification effect of the model, as shown in the right-bottom box in Fig. 2.

In this context, the main differences between our work and the other related work are given as follows.

1) We construct a unique lightweight subnetwork composed of 1-D and 2-D convolution, whereas the existing related methods are mainly based on 3-D architecture, which are deeper and more complex and thus easily get trapped into overfitting under limited training samples.
2) We design a differentiated input strategy, which embeds patches with different sizes into the dual branches of our Siamese network and thus provides more features to the model, whereas most of the exiting related methods used the same patch size.
3) We propose a weighted contrast loss and an adaptive cross-entropy loss, which can improve the discrimination power of our method, whereas existing related methods rarely exploited the loss functions.

III. Proposed Method

We first build a lightweight network (SSN) with few parameters to extract spectral–spatial information of HSI data. Then, we construct a Siamese network framework (S3Net) based on SSN. Finally, a weighted contrastive loss and an adaptive cross-entropy loss are designed. The training and test mechanisms of S3Net are shown in Fig. 3. In data

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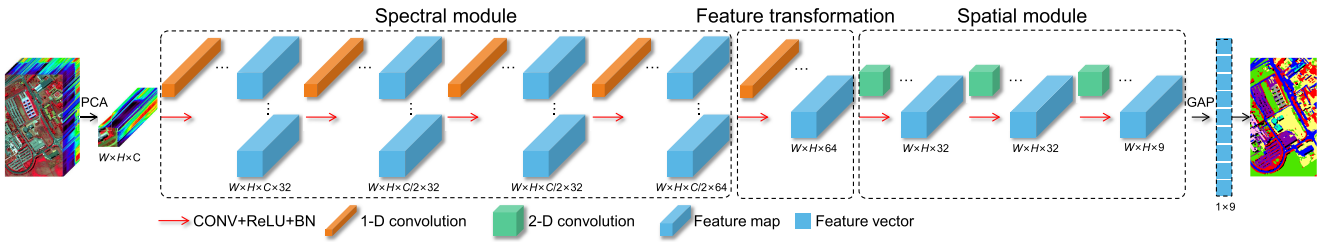


Fig. 4. Architecture of SSN.

TABLE I
ARCHITECTURE OF SSN

	Convolution	ReLU	BN	Output shape	Filter size	Padding
Input	–	–	–	$W \times H \times C$	–	N
Spectral module	1	Y	$1 \times C$	$W \times H \times C \times 32$	$1 \times 1 \times 3 \times 1 \times 32$	Y
	2	Y	$1 \times C/2$	$W \times H \times C/2 \times 32$	$1 \times 1 \times 2 \times 32 \times 32$	N
	3	Y	$1 \times C/2$	$W \times H \times C/2 \times 32$	$1 \times 1 \times 3 \times 32 \times 32$	Y
	4	Y	$1 \times C/2$	$W \times H \times C/2 \times 64$	$1 \times 1 \times 3 \times 32 \times 64$	Y
Feature transformation	5	Y	1×64	$W \times H \times 64$	$1 \times 1 \times C/2 \times 64 \times 64$	N
Spatial module	6	Y	1×32	$W \times H \times 32$	$3 \times 3 \times 64 \times 32$	Y
	7	Y	1×32	$W \times H \times 32$	$3 \times 3 \times 32 \times 32$	Y
	8	Y	$1 \times K$	$W \times H \times K$	$3 \times 3 \times 32 \times K$	Y
GAP	–	–	–	$1 \times 1 \times K$	–	N

preprocessing, samples will be divided into the training and test subsets, where each sample will be generated into two different sizes of patches. During the training stage, each large patch sample will be paired with all small patch samples, and paired samples will be fed into the two branches of S3Net to calculate the weighted contrastive loss. In addition, if sample pairs centered on the same pixel, they will be used to calculate additional adaptive cross-entropy loss. During the testing stage, each large patch sample will be paired with its small patch sample, and their features obtained by two branches will be added for predicting.

A. Spatial–Spectral Network

We design a lightweight SSN, which is composed of 1-D and 2-D convolutions. In this case, network parameters will be reduced by 2/3 compared with 3-D convolution, which is also beneficial to weaken the overfitting issue under small samples. The architecture of SSN is shown in Fig. 4 and Table I reports the specific dimension of different layers of SSN, where C is the dimension of patch processed after conducting principal component analysis (PCA), K is the number of classes, and $W \times H$ represents the patch size, with W and H the width and height of the patch, respectively.

1) Spectral Module: This module is designed to extract the local spectral feature of input by 1-D convolution. The input $X \in \mathbb{R}^{W \times H \times C}$ passes through four 1-D convolution layers, where ReLU and BN are connected after each convolution layer. We set the step size of the second convolution layer to 2 for reducing the feature map dimension. After four convolution layers, the dimension of feature map is $X \in \mathbb{R}^{W \times H \times C/2 \times 64}$ (64 feature maps with dimension of $W \times H \times C/2$). The spectral module extracts the local spectral features through four 1-D convolution layers and achieves dimensional reduction, which further reduces the overall parameters of SSN. In addition, the 1-D convolution used in this article is implemented by setting the spatial size of 3-D convolution to 1 as shown in the following equation:

$$v_{ij}^{xyz} = f\left(\sum_{m}\sum_{p=1}^{1}\sum_{q=1}^{1}\sum_{r=0}^{R_i-1} w_{ijm}^{pqr} v_{(i-1)m}^{xy(z+r)} + b_{ij}\right)$$
$$f(x) = \max(x, 0) \tag{2}$$

where v_{ij}^{xyz} represents the value of a neuron at position (x, y, z) of the jth feature map in the ith layer. The parameter m indexes the feature map in the $(i-1)$th layer connected to the current (jth) feature map, and p and q are the height and the width of the spatial convolution kernel (where p and q are set to 1), respectively. The parameter R_i is the size of the kernel along spectral dimension, w_{im}^{pqr} is the value of position (p, q, r) connected to the mth feature map, and b_{ij} is the bias of the jth feature map in the ith layer. In addition, $f(x)$ is a ReLU activation function as shown in (2).

The number of parameters of the model should not be too large under small samples, and therefore, we use PCA and one convolution layer to reduce the spectral bands, thus reducing the model parameters. However, reducing the number of spectral bands using convolution will lead to information loss. In order to ensure the number of features, this article only uses one layer of convolution in the network for dimensionality reduction.

2) Feature Transformation: This module integrates the channel information obtained by the spectral module into feature map. We use a 1-D convolution to extract the global spectral information, where ReLU and BN layers are connected, and the obtained output $X \in \mathbb{R}^{W \times H \times 1 \times 64}$ is reorganized into $X \in \mathbb{R}^{W \times H \times 64}$. The value of a neuron v_{ij}^{xy} at position (x, y) of

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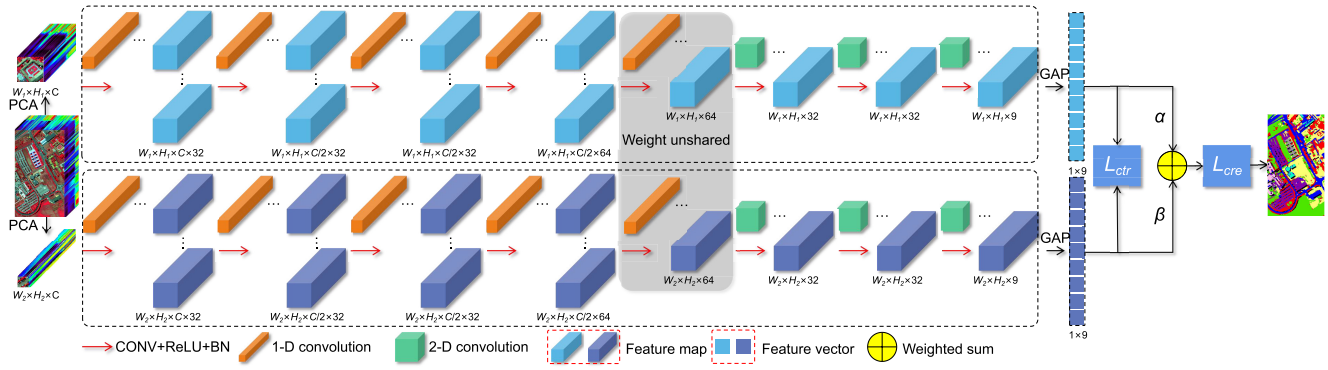


Fig. 5. Architecture of S3Net.

the jth feature map in the ith layer (in this module, i represents the layer of feature transformation) is given in the following equation:

$$v_{ij}^{xy} = f\left(\sum_{m}\sum_{p=1}^{1}\sum_{q=1}^{1}\sum_{r=1}^{C/2} w_{ijm}^{pqr} v_{(i-1)m}^{xyr} + b_{ij}\right) \tag{3}$$

where m indexes the feature map layer in the previous layer connected to the current layer and p and q are the height and the width of the spatial convolution kernel (p and q set to 1 to implement 1-D convolution in feature transformation), respectively. The kernel size r is equal to the input channel dimension$(C/2)$, w_{ijm}^{pqr} is the value of position (p, q, r) connected to the mth feature map, and b_{ij} is the bias of the jth feature map in the ith layer. The kernel size of convolution in this part is equal to the channel dimension of input, which means that this convolution layer can extract global spectral information and realize full connection in spectral space.

3) Spatial Module: This module extracts spatial information of the feature map obtained by the feature transformation module. The feature map $X \in \mathbb{R}^{W\times H\times 64}$ passes through three 2-D convolution layers to extract spatial information, where ReLU and BN are connected after each layer, and the feature map dimension will be reduced from 64 to K. After this module, the feature map becomes $X \in \mathbb{R}^{W\times H\times K}$, and each feature map represents a class. The value of a neuron v_{ij}^{xy} at position (x, y) of the jth feature map in the ith layer is given in the following equation:

$$v_{ij}^{xy} = f\left(\sum_{m}\sum_{p=0}^{P_i-1}\sum_{q=0}^{Q_i-1} w_{ijm}^{pq} v_{(i-1)m}^{(x+p)(y+q)} + b_{ij}\right) \tag{4}$$

where m indexes the feature map in the $(i-1)$th layer connected to the current (jth) feature map, w_{ijm}^{pq} is the weight of position (p, q) connected to the mth feature map, P_i and Q_i, respectively, denote the height and width of the spatial convolution kernel, and b_{ij} is the bias of the jth feature map in the ith layer.

We use padding in the whole spatial module. It is because after three 2-D convolution layers, the receptive field will be the same to the small patch size in S3Net, and thus, there is only one feature vector left, which represents the global features without padding. However, if we use padding, the center feature vector is still obtained by the whole patch, and for the edge feature vectors, they are not formed by whole pixels; therefore, some local features can be retained [42]. In this context, we think that the feature map can retain global and local features using padding. In addition, the padding can make edge pixels participate in the convolution multiple times, and thus, the network will not lose the boundary information.

4) Classification: The feature map obtained by spatial module has included the class information, which means that different layers of feature map represent different classes. We use GAP instead of fully connected layer to get feature vector, as shown in (5), which can further reduce the complexity of the model

$$X_{\text{out}} = \sum_{i=1}^{W}\sum_{j=1}^{H} X_{ijK}/(W * H) \tag{5}$$

where X_{ijK} is the output of spatial module and i and j represent the windows size indexes. After GAP, the feature map is compressed to $X \in \mathbb{R}^{1\times K}$. Softmax is used to obtain the class probability by calculating the cross-entropy loss

$$L_{\text{cre}}(\widehat{y}_i, y_i) = -\sum_{i=0}^{N} y_i \log(\widehat{y}_i) \tag{6}$$

where N is the total number of training samples, $\widehat{y}_i$ is the class probability of the ith sample, and y_i is the one-hot label of the ith sample.

B. S3Net

Based on two SSNs, we construct a Siamese network framework, i.e., S3Net whose model architecture is shown in Fig. 5, where $W_1 \times H_1$ represents the big patch and $W_2 \times H_2$ represents the small patch. In this framework, samples are expanded by pairing and the separability is enhanced through metric learning. First, HSI will be generated into two different size patches, which are then fed into two SSNs to get feature vectors. Second, the feature vectors will be used to calculate weighted contrastive loss and adaptive cross-entropy loss for training the model. Finally, the label of test sample will be predicted by adding the two feature vectors.

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TABLE II
COMPARISON OF SAMPLE COMPOSITION UNDER DIFFERENT NETWORK STRUCTURES

	Same input		Different input	
	Classify	Metric	Classify	Metric
Positive	$K\times T$	$K\times T\times(T\text{-}1)/2$	$K\times T$	$(K\times T)\times(T\text{-}1)$
Negative	–	$K\times T\times(K\text{-}1)\times T/2$	–	$(K\times T)\times(K\text{-}1)\times T$

1) Differentiated Input: We use two different size inputs for two branches in S3Net, which is different from the traditional Siamese network that are with the same size inputs. In this strategy, each pixel generated two size patch samples and each sample pair is composed of a large and a small patch sample. Different samples have different feature distributions, which means that they can provide more features to the model for strengthening the recognition ability. In order to improve the adaptability of the two subnetworks for different inputs, we use unshared weights in the feature transformation module and shared weights in the other layers (see Fig. 5).

2) RS for Negative Pairs: Siamese network realizes data augmentation through sample pairing in the training process. When using the same input, only one sample pair can be formed using two samples. On the contrary, two size combination sample pairs can be formed with the same two samples in a differentiated input strategy. However, there is a serious imbalance issue in the ratio of positive and negative samples for Siamese network framework. The specific sample composition is shown in Table II, where T represents the number of samples per class. The result shows that the ratio of positive and negative samples is approximately $1{:}(K-1)$. Too many negative samples will consume a lot of training time and even deteriorate the network. To increase the converging speed and weaken the overfitting issue, we randomly select the negative samples equal to positive samples in each epoch without replacement.

C. Loss Functions

In the framework of S3Net, there are two loss functions in each training epoch. The final loss is given as

$$L = L_{\text{ctr}} + L_{\text{cre}}. \tag{7}$$

An Adam optimizer [43] is used to optimize the network with L.

1) Weighted Contrastive Loss: Positive samples with large distance and negative samples with small distance are regarded as hard samples for Siamese network, which are more important in training. However, the label of sample pairs in the traditional Siamese network is stable. In order to increase the proportion of hard samples in training, we design weight for contrastive loss to focus on these hard samples.

We take center vectors $L_1 \in \mathbb{R}^{1\times K}$ and $L_2 \in \mathbb{R}^{1\times K}$ from the two feature maps $X_1 \in \mathbb{R}^{P_1\times P_1\times K}$ and $X_2 \in \mathbb{R}^{P_2\times P_2\times K}$ (see Fig. 6, where P_1 and P_2 represent the size of the two feature maps, respectively) to calculate the cosine distance w between sample pairs as

$$w = \frac{L_1 \bullet L_2}{\|L_1\|\|L_2\|}. \tag{8}$$

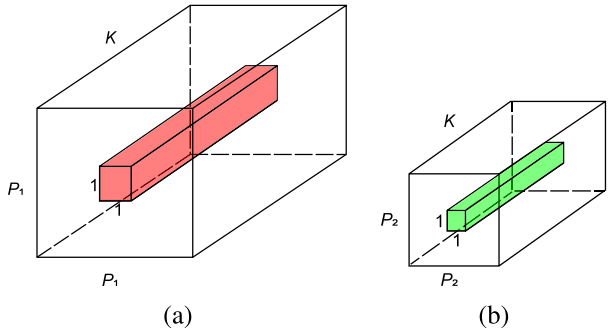


Fig. 6. Feature vector of the center pixel within two patch sizes. (a) Large patch, where the red rectangle is the feature vector. (b) Small patch, where the green rectangle is the feature vector.

Then, we multiply $(1-w)$ and w to positive and negative samples, respectively, as shown in the following equation:

$$L_{\text{ctr}}(y, d_{\text{pos}}, d_{\text{neg}}) = y\times(1-w)\times d_{\text{pos}} + (1-y)\times w\times\max(m-d_{\text{neg}}, 0) \tag{9}$$

where the distance between sample features is d_{pos} when the two samples are belonging to the same class and the distance is d_{neg} when the two samples are from different classes.

The weighted contrastive loss can pay more attention to hard samples through the design of w, thereby prompting the model to fit in the correct direction.

2) Adaptive Cross-Entropy Loss: To balance the information due to differentiated inputs, we fuse the two feature vectors learned from the dual branches using two adaptive parameters (α and β). Then, an adaptive cross-entropy loss can be defined with the fused vector to train the network

$$L_{\text{cre}}(\hat{y}_i, y_{iP_1+iP_2}) = -\sum_{i=0}^{N} y_{iP_1+iP_2}\log(\widehat{y_i})$$
$$y_{iP_1+iP_2} = \text{Softmax}(\alpha\times y_{iP_1} + \beta\times y_{iP_2}) \tag{10}$$

where y_{iP_1} and y_{iP_2} are the two feature vectors for the ith sample. Two parameters α and β are $1\times K$-dimensional vectors with an initial value of 0.5, which are learned in the training process. The pseudocode of the proposed S3Net method is shown in Algorithm 1.

IV. EXPERIMENTAL RESULTS

A. Hyperspectral Datasets

1) The first HSI was acquired by the AVIRIS sensor over the Indian Pines region in Northwestern Indiana in 1992. The image size in pixel is 145 × 145, with a moderate spatial resolution of 20 m. The number of data channels in the acquired image is 220 (with a spectral range from 0.4 to 2.5 μm). A total of 200 radiance channels are used in the experiments by removing several noisy and water-absorbed bands. A three-band false-color composite image and the ground-truth map are shown in Fig. 7. As reported in Table III, a total of 10 249 samples containing 16 classes are available.

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Algorithm 1 S3Net

Input: HSI patches X_1 and X_2, reduced dimension C, total epochs N_{epo}.
I. Training stage:
Split all labeled data into training and test sets.
Pair each large patch with all small patch to obtain sample pairs $\{i_{P_1}, i_{P_2}\}$ $(i \in N_{\text{Train}})$.
for epoch from 1 to N_{epo} **do**
    Feed $\{i_{P_1}, i_{P_2}\}$ into S3Net to get feature vectors y_{iP_1} and y_{iP_2}.
    Calculate L_{ctr} using Eq. (9) with y_{iP_1} and y_{iP_2}.
    if $\{i_{P_1}, i_{P_2}\}$ are centered on the same pixel **then**
        Calculate L_{cre} using Eq. (10) with y_{iP_1} and y_{iP_2}.
        Train the network using L_{ctr} and L_{cre} with Adam optimizer.
    end if
end for
II. Test stage:
Pair each large patch with all small patch to obtain sample pairs $\{i_{P_1}, i_{P_2}\}$ $(i \in N_{\text{Test}})$.
for i from 1 to N_{Test} **do**
    Feed $\{i_{P_1}, i_{P_2}\}$ into S3Net to get feature vectors y_{iP_1} and y_{iP_2}.
    Get the predicted label using arg max with $(\alpha \times y_{iP_1} + \beta \times y_{iP_2})$.
end for
Output: Predicted labels for the whole image.

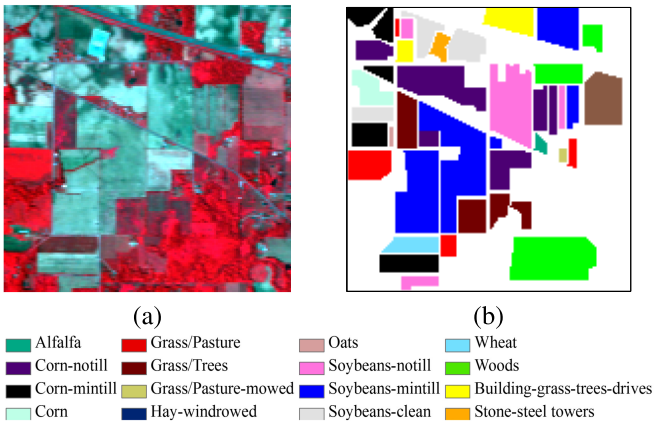


Fig. 7. AVIRIS Indian Pines datasets. (a) False-color composite image (R: 50, G: 27, and B: 17). (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

2) The second HSI was acquired by the ROSIS sensor over the urban area of the University of Pavia, Italy. The image size in pixel is 610 × 340, with a very high spatial resolution of 1.3 m. The number of data channels in the acquired image is 103 (with a spectral range from 0.43 to 0.86 μm). A three-band false-color composite image and the ground-truth map are shown in Fig. 8. As reported in Table IV, a total of 42 776 samples containing nine classes are available.
3) The third HSI was acquired by the AVIRIS sensor over Salinas Valley in southern California, USA.

TABLE III
NUMBER OF LABELED SAMPLES PER CLASS FOR THE INDIAN PINES DATASETS

Label	Class	# Number
1	Alfalfa	40
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass/Pasture	483
6	Grass/Trees	730
7	Grass/Pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybeans-notill	972
11	Soybeans-mintill	2455
12	Soybeans-clean	593
13	Wheat	205
14	Woods	1265
15	Bldg-grass-tree-drives	386
16	Stone-steel towers	93
Total (16 classes)		10249

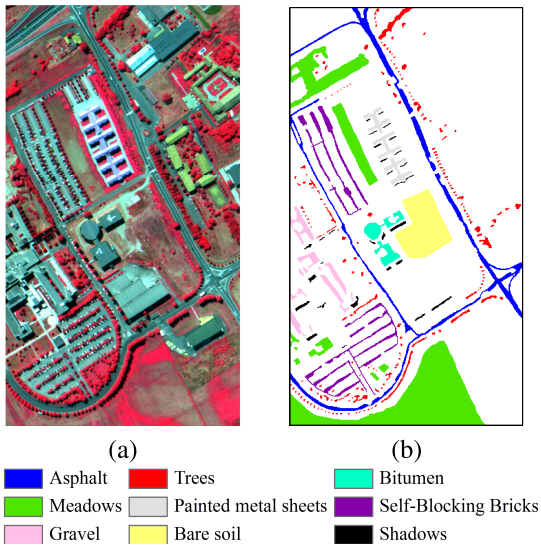


Fig. 8. ROSIS University of Pavia datasets. (a) False-color composite image (R: 102, G: 56, and B: 31). (b) Ground-truth map containing nine mutually exclusive land-cover classes.

TABLE IV
NUMBER OF LABELED SAMPLES PER CLASS FOR THE UNIVERSITY OF PAVIA DATASETS

Label	Class	# Number
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
Total (9 classes)		42776

The image size in pixels is 512 × 217, with a spatial resolution of 3.7 m. The number of data channels in the acquired image is 224 (with a spectral range from 0.4 to 2.5 μm). A total of 204 radiance channels are used in the

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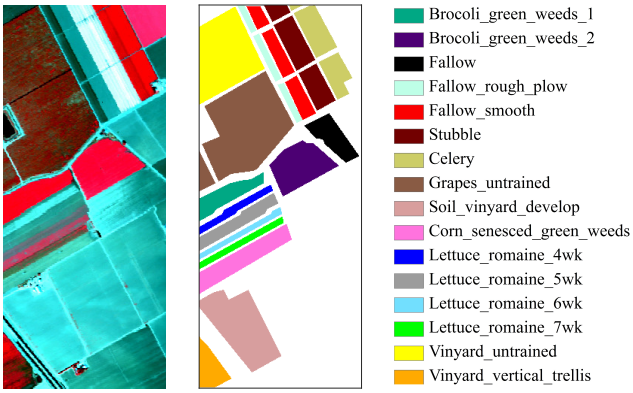


Fig. 9. AVIRIS Salinas datasets. (a) False-color composite image (R: 57, G: 27, and B: 17). (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

TABLE V
NUMBER OF LABELED SAMPLES PER CLASS FOR THE SALINAS DATASETS

Label	Class	# Number
1	Broccoli_green_weeds_1	2009
2	Broccoli_green_weeds_22	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vineyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vineyard_untrained	7268
16	Vineyard_vertical_trellis	1807
Total (16 classes)		54129

experiments by removing the noisy and water-absorbed bands. A three-band false-color composite image and the ground-truth map are shown in Fig. 9. As reported in Table V, a total of 54 129 samples containing 16 classes are available.

B. Experimental Settings

The following conditions hold.

1) *For Hyperparameters:* Reduced dimension after PCA, a number of training epochs, patch sizes, and learning rates are analyzed in the flowing experiments of SSN. Margin parameter (m) and positive and negative samples ratio are analyzed in the flowing experiments of S3Net.
2) *Comparison With SSN:* For comparing the performance of SSN under small training samples, we consider six methods, including SVM [44], RF [45], 2-D CNN [20], 3-D CNN [46], spectral–spatial unified network (SSUN) [47], and spectral–spatial residual network (SSRN) [48]. SVM and RF are traditional methods, 2-D CNN and 3-D CNN are classic DL methods, and SSUN and SSRN are start-of-the-art DL methods, whose parameters are determined according to the corresponding recommended values.
3) *Generalization Performance of S3Net:* For comparing the generalization performance of S3Net under different samples, we consider seven methods, including 2-D CNN, 3-D CNN, SSUN, SSRN, 3-D autoencoder and Siamese network for semisupervised HSI classification (3DAES) [36], deep cross-domain few-shot learning (DCFSL) [39], and a Siamese network composed of double SSN (D-SSN). 3DAES is proposed for limited training samples with a semisupervised Siamese network combined with 3-D autoencoder, and DCFSL is a transfer learning method for limited labeled samples, whose parameters are determined according to the corresponding recommended values. Moreover, to verify the effect of the improved strategies in this article, we compare S3Net with D-SSN, which is constructed by two SSNs directly in a Siamese network framework without differentiated input, weighted contrastive loss, and adaptive cross-entropy loss. We test these methods under {5, 10, 15, 20, 30, 40, 50} training samples to evaluate the classification ability and the generalization performance of the model under small samples. It is worth noting that there are no labeled samples for validation, and all the rest samples are used for test. In addition, FSL methods mainly focus on using small samples for training [23], [36].
4) *Evaluation Metrics:* The overall accuracy (OA), average accuracy (AA), and Kappa coefficient are used to evaluate the performance of different models, and we report the averaged results after conducting ten independent runs in all experiments.
5) *Running Platform:* All experiments are carried out under Windows 10, Intel i7-9700 k, NVIDIA GeForce RTX 2060 SUPER, and 64-GB memory hardware. Different DL frameworks are built on Pytorch 1.14 and TensorFlow 1.14 in the Python 3.6 environment. The source codes will be available online.[1]

C. Hyperparameter Analysis for SSN

To analyze the impacts of learning rate of SSN on OA, we set the learning rate from the range of {0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001} for the three datasets and the evolution of OA along with learning rate is shown in Fig. 10(a). As we can see, when the learning rate is set to 0.01, OA is generally higher than others. Therefore, we set it to 0.01 in the subsequent experiments.

As for the patch size, we set its range of {3 × 3, 5 × 5, 7 × 7, 9 × 9, 11 × 11, 13 × 13, 15 × 15, 17 × 17} for the three datasets and the evolution of OA along with patch size is shown in Fig. 10(b). As we can see, when the patch size is set to 9 × 9, OA is generally higher than others. Therefore, we set it to 9 × 9 in the subsequent experiments.

To analyze the impacts of the number of PCs retained on OA, we set the number of PCs to the range of {10, 20, …, 100, 103} in the University of Pavia, {10, 20, …, 190, 200} in Indian Pines, and {10, 20, …, 200, 204} in Salinas. The evolution of OA along with the retained PCs is

[1]https://github.com/ZhaohuiXue/S3Net

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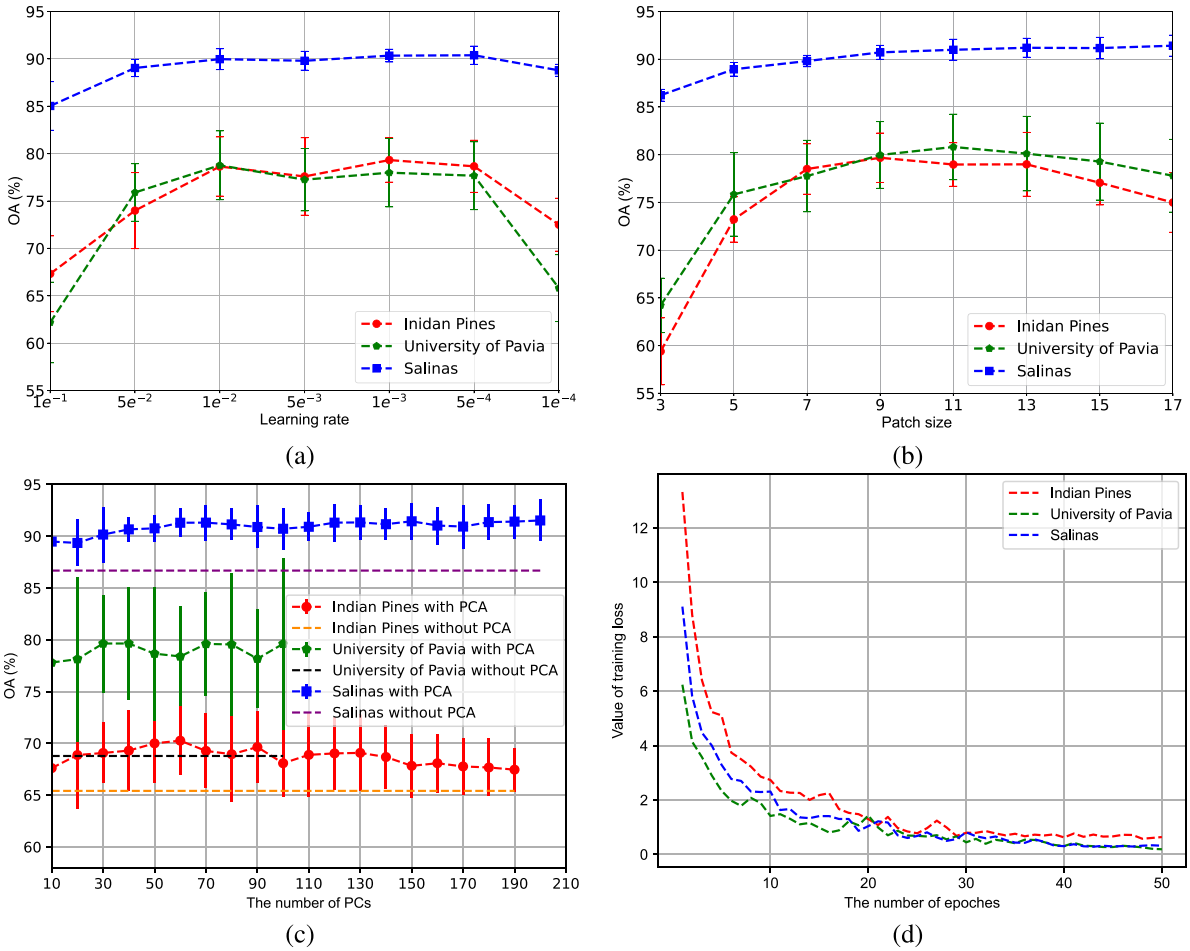


Fig. 10. Evolution of OA as a function of (a) learning rate, (b) patch size, (c) number of PCs. Evolution of training loss as a function of (d) number of epochs (five labeled samples per class). Error bars represent standard deviations.

shown in Fig. 10(c). As we can see, for the three datasets, the OAs without using PCA are lower than that of using PCA. Specifically, for Indian Pines, when the number of PCs is 60, the model can get the highest OA, and we then set the number of PCs to 60. For the University of Pavia, the highest OA is achieved when the number of PCs is 30, and thus, we set the number of PCs to 30. For Salinas, when the number of PCs is greater than 60, the OA is relatively high, and we set the number of PCs to 60.

To analyze the impacts of epoch on training loss, we set the epoch to the range of 0–50 for the three datasets. The evolution of training loss along with epoch is shown in Fig. 10(d). As we can see, the value of training loss is becoming steady on the three datasets when the number of epochs reaches 30. Therefore, to ensure adequate training and avoid overfitting, we set it to 30 in the subsequent experiments.

D. Classification Results for SSN

To validate the superiority of the proposed SSN, we compare SVM, RF, 2-D CNN, 3-D CNN, SSUN, SSRN, and SSN using five labeled samples per class in three datasets, and the averaged results are reported in Table VI. According to the results, SSN outperforms the other counterparts with the highest accuracies. Specifically, SSN obtains an OA of 70.26%, which yields 3.1%–29.86% improvements compared to the other methods for the Indian Pines dataset. For the University of Pavia datasets, SSN obtains the highest OA of 79.03%, which is 1.56%–17.39% higher than the other methods. As for the Salinas datasets, our method produces an OA of 91.62%, which is 2.43%–30.53% higher than the other methods. For AA and Kappa, SSN also produces significant improvements compared to others, except that AA obtained by SSRN in the University of Pavia is higher than SSN. The results demonstrate that our method is suitable for small sample problems by using 1-D and 2-D convolutions to extract spectral–spatial features and GAP to classify, which greatly reduces the number of network parameters and weakens the overfitting issue.

E. Ablation Analysis of S3Net

1) Differentiated Input: To verify the effectiveness of differentiated input strategy, we set the two sizes of sample pair to the range of {3, 5, 7, 9, 11, 13} with five training samples per class, and the averaged OAs are shown in Fig. 11. According to the results, the differentiated input strategy gets higher accuracies than using sample pairs with the same size

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TABLE VI
CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE THREE DATASETS (FIVE LABELED SAMPLES PER CLASS)

Data sets	Metrics	SVM [44]	RF [45]	2-D CNN [20]	3-D CNN [46]	SSUN [47]	SSRN [48]	SSN (Ours)
Indian Pines	OA (%)	40.40 ± 3.36	43.45 ± 2.82	57.95 ± 5.89	63.75 ± 3.99	59.02 ± 1.82	67.16 ± 4.43	**70.26 ± 3.66**
	AA (%)	52.28 ± 2.15	55.06 ± 2.15	72.20 ± 2.94	76.74 ± 2.85	74.16 ± 1.35	69.04 ± 4.94	**82.17 ± 2.01**
	Kappa×100	34.06 ± 3.29	37.15 ± 2.97	53.32 ± 6.02	59.52 ± 4.21	54.23 ± 1.85	63.34 ± 4.71	**66.65 ± 3.94**
University of Pavia	OA (%)	61.64 5.50	58.42 ± 6.03	70.13 ± 7.96	73.73 ± 7.05	73.82 ± 6.00	77.47 ± 5.28	**79.03 ± 5.94**
	AA (%)	72.79 ± 2.10	70.54 ± 2.21	80.09 ± 2.54	79.90 ± 3.76	75.40 ± 3.37	**87.20 ± 2.53**	85.08 ± 3.97
	Kappa×100	53.10 ± 5.18	49.31 ± 5.05	63.13 ± 8.11	66.93 ± 8.32	66.71 ± 6.32	71.95 ± 6.00	**73.44 ± 6.92**
Salinas	OA (%)	61.09 ± 0.01	82.67 ± 1.79	88.45 ± 1.80	86.96 ± 4.13	85.36 ± 1.81	89.19 ± 1.87	**91.62 ± 2.17**
	AA (%)	65.18 ± 1.17	89.06 ± 0.92	93.16 ± 1.07	91.05 ± 4.36	89.73 ± 1.11	92.64 ± 1.26	**95.70 ± 1.19**
	Kappa×100	57.12 ± 1.17	80.77 ± 1.92	87.18 ± 1.95	85.56 ± 4.53	83.80 ± 1.99	87.95 ± 2.12	**90.70 ± 2.38**

when the patch sizes of the two branches are both greater than 5. Specifically, for University of Pavia in Fig. 11(b), the three patch size combinations of {11,7},[2] {9,7}, and {13,7} have achieved the highest accuracy. As for the other two datasets, the patch size combination of {13,7} has also achieved higher accuracy. In this context, we set that the optimal patch size combination is {13,7} in the subsequent experiments. This differentiated input strategy can alleviate the spatial autocorrelation issue to some extent.

To exhibit the distribution differences between two differentiated inputs, we plot the value distribution of two size samples for Corn-mintill and Bldg-grass-tree-drives in Indian Pines, as shown in Fig. 12. For large patch samples of the two classes, values near 0 are significantly higher, while the small patches of the two classes are smoother. The above experimental results illustrate that a differentiated input strategy can provide more features, and thereby, the model can improve the recognition ability of producing higher classification accuracy with more features.

2) RS Strategy: To analyze the impacts of the positive and negative sample ratio on OA, we fix the total sample size of each batch and change the positive and negative sample ratio as {4:1, 3:1, 2:1, 1:1, 1:2, 1:3, 1:4} for the three datasets. The evolution of OA along with positive and negative ratios is shown in Fig. 13. As we can see, for Indian Pines and University of Pavia, the two OA curves are very similar, and when positive samples are equal to negative samples, the model can get the highest OA. For Salinas, because of the high homogeneity within class and separability between class, the model always gets relatively high OAs under different cases. Therefore, we set the positive and negative ratios to 1:1 for the three datasets.

To verify the effectiveness of RS strategy in three datasets, we compare different selection strategies under five samples per class, and the averaged OA, training time, and test time are reported in Table VII. As we can see, the RS strategy slightly increases the accuracy but greatly reduces the training time compared without using RS. For OA, we consider that the RS strategy can effectively balance the unbalance amount between positive and negative samples, but it does not generate new data to improve the feature extraction ability. Therefore, it does not show obvious advantages in improving OA. For training time, the RS strategy uses fewer training samples,

[2]We denote by $\{a, b\}$ the combination of two patches with size $a \times a$ and $b \times b$.

TABLE VII
IMPACT OF RS FOR SAMPLE PAIRING IN THE THREE DATASETS (FIVE LABELED SAMPLES PER CLASS)

Data	RS or not‡	OA (%)	Training time (s)	Test time (s)
IP†	N	76.15±2.56	207.10±2.31	7.55±0.24
	Y	**76.35±2.99**	**4.65±0.21**	**7.40±0.13**
PU†	N	84.54±4.41	23.33±1.21	22.23±0.11
	Y	**84.62±4.08**	**2.60±0.3 0**	**22.45±0.12**
SA†	N	93.10±1.38	178.66±1.87	37.70±0.32
	Y	**93.18±1.43**	**4.69±0.24**	**37.67±0.29**

†: IP for Indian Pines, PU for University of Pavia, and SA for Salinas.
‡: Y for using random selection for sample pairing, and N for not.

and thus, the training time in three datasets is reduced a lot. The test time of the two strategies is almost the same since the number of test sample pairs is the same.

3) Loss Function: To choose the best m value in contrastive loss, we optimize the value of m from the range of {0.1, 0.2, …, 2.0} for the three datasets and the evolution of OA along with m is shown in Fig. 14. As we can see, when m is in the range of 1.1–1.4, the model gets higher OAs on the three datasets. The parameter m controls the distance between positive and negative samples, and in order to better distinguish positive and negative samples, we set it to 1.1 in the subsequent experiments.

To verify the effectiveness of weighted contrastive loss, we compare traditional contrastive loss and weighted contrastive loss in three datasets with five samples per class, and the averaged OAs are reported in Table VIII. Compared with traditional contrastive loss, weighted contrastive loss has better accuracy in three datasets with increased OA of 0.12%-1.31%. The results demonstrate that the weight obtained by center vectors of feature maps from the two branches can truly reflect the similarity of the sample pairs, thereby improving the converging performance with enhanced recognition ability for hard samples. As for the effectiveness of adaptive cross-entropy loss, we compare traditional cross-entropy loss and adaptive cross-entropy loss in three datasets with five samples per class. According to the results reported in Table VIII, adaptive cross-entropy loss gets the highest OA in three datasets, which is because the loss can get an appropriate ratio for the two branches to better express feature by training the adaptive parameters.

F. Generalization Performance of S3Net

To validate the generalization performance of the proposed S3Net, we compare the classification results of different

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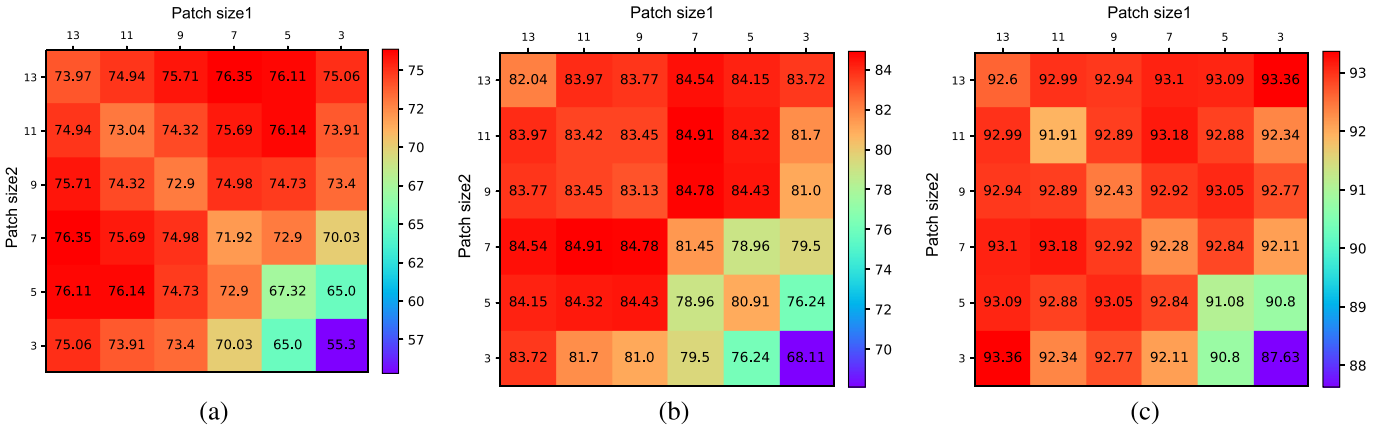


Fig. 11. Impact of differentiated input patch size on OA for the three datasets (five labeled samples per class). (a) Indian Pines. (b) University of Pavia. (c) Salinas.

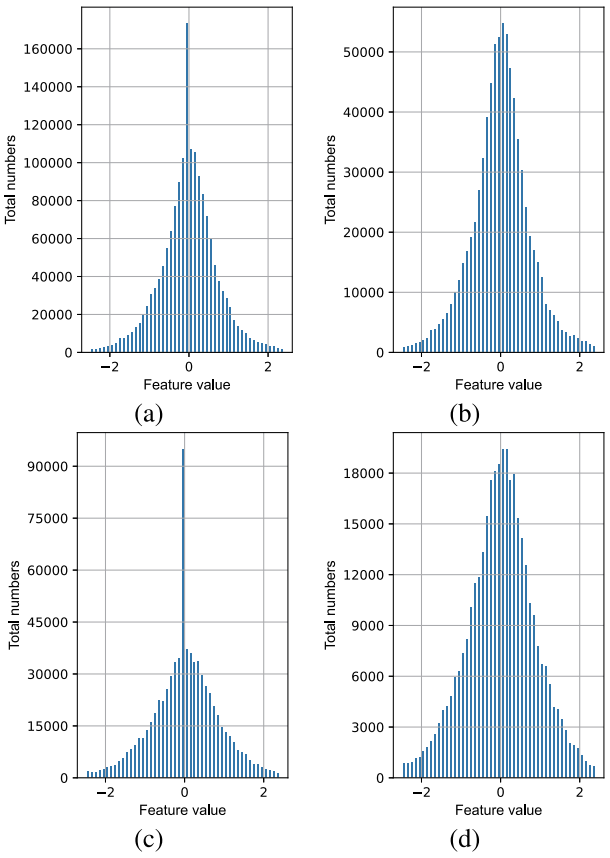


Fig. 12. Feature distribution of different patch sizes in Indian Pines for some chosen classes. (a) Corn-mintill with 13×13 patch size. (b) Corn-mintill with 7×7 patch size. (c) Bldg-grass-tree-drivers with 13×13 patch size. (d) Bldg-grass-tree-drivers with 7×7 patch size.

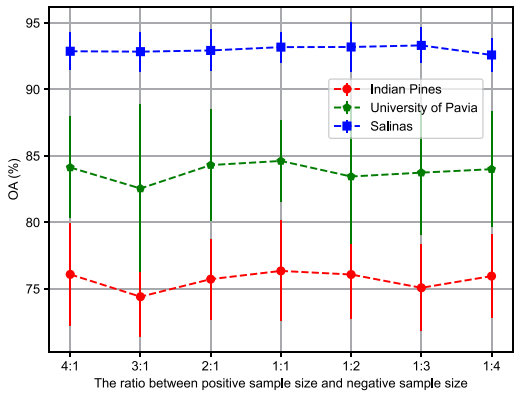


Fig. 13. Evolution of OA as a function of the ratio between positive and negative sample size (five labeled samples per class). Error bars represent standard deviations.

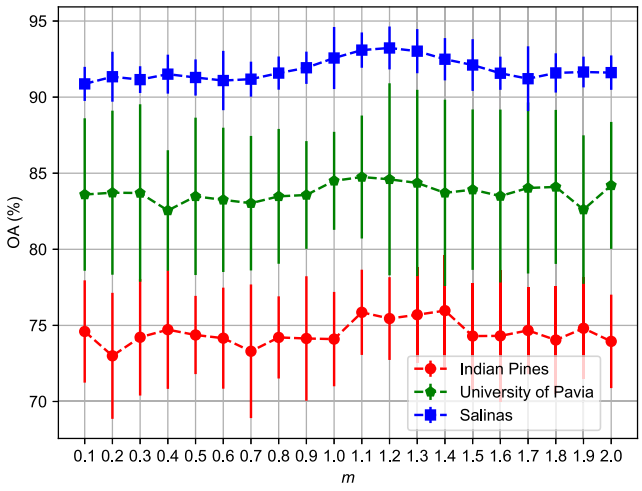


Fig. 14. Evolution of OA as a function of m (five labeled samples per class). Error bars represent standard deviations.

methods under few training samples ranging from 5, 10, 15, 20, 30, 40, 50, and the evolution of OA as a function of different number of training samples is given in Fig. 15. As we can see, S3Net gets the highest accuracies in different cases in three datasets, which proves the superiority in terms of generalization performance of our method. We consider that both the differentiated input strategy and the designed two losses are beneficial to the performance improvements.

G. Classification Results for S3Net

To validate the superiority of the proposed S3Net, we compare 2-D CNN, 3-D CNN, SSUN, SSRN, 3DAES, DCFSL, D-SSN, and S3Net using 20 labeled samples per class.

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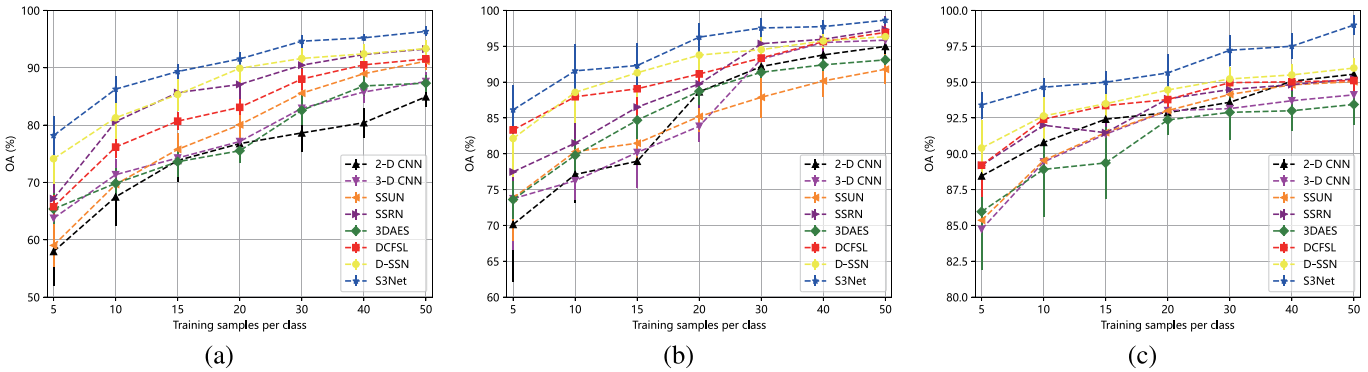


Fig. 15. Evolution of OA as a function of number of training samples per class. (a) Indian Pines. (b) University of Pavia. (c) Salinas.

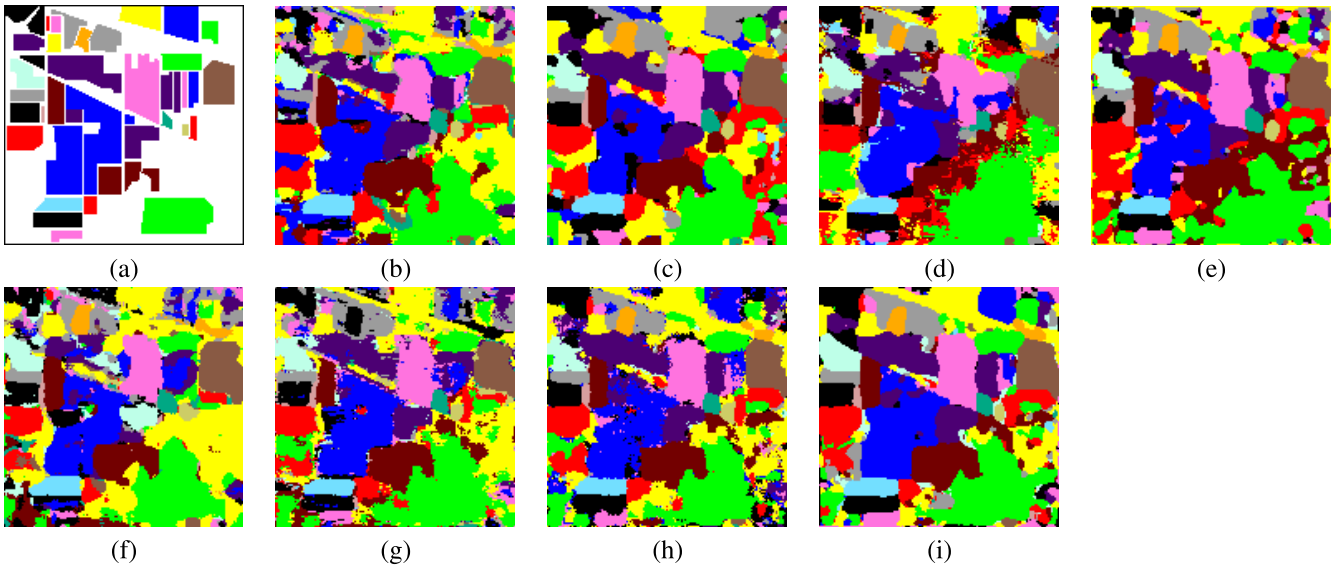


Fig. 16. Classification maps obtained by different methods for the Indian Pines dataset (20 labeled samples per class). (a) Ground truth. (b) 2-D CNN (76.77%). (c) 3-D CNN (77.13%). (d) SSUN (80.05%). (e) SSRN (87.09%). (f) 3DAES (75.52%). (g) DCFSL (83.10%). (h) D-SSN (89.91%). (i) S3Net (91.54%).

TABLE VIII

COMPARISON OF DIFFERENT LOSS FOR THE THREE DATASETS (FIVE LABELED SAMPLES PER CLASS)

Loss	Indian Pines	University of Pavia	Salinas
L_{ctr}	76.15±2.99	84.62±4.08	93.08±1.43
Weighted L_{ctr}	**77.46±2.54**	**85.34±3.75**	**93.20±1.12**
L_{cre}	77.46±2.54	85.04±3.75	93.20±1.12
Adaptive L_{cre}	**78.18±3.33**	**86.13±3.41**	**93.40±0.82**

According to the result of Indian Pines reported in Table IX, S3Net and D-SSN outperform the other methods obviously with the highest accuracies. Specifically, S3Net obtains the highest OA of 91.54%, which achieves significant improvement, ranging from 16.02% to 1.63%, which is higher than the other methods. In addition, S3Net also produces significant improvements in terms of AA and Kappa compared to others. As for the class-specific accuracy, S3Net gets the highest accuracies in 10 of 16 classes, which also proves the superiority of S3Net.

As shown in Fig. 16, S3Net produces more smooth classification map with less noise, which gets higher homogeneity within class and better separability between class. In particular, S3Net obtains the smoothest results for woods and soybeans-notil classes, whereas the other methods have many missclassified pixels, which is in accordance with the results reported in Table IX. In the unlabeled area, S3Net can also accurately classify the test samples.

According to the result of University of Pavia reported in Table X, S3Net and D-SSN outperform the other methods with the highest accuracy. Specifically, S3Net obtains the highest OA of 96.27%, which is 12.43%–2.50% higher than the other methods. In addition, S3Net also produces significant improvements in terms of AA and Kappa. As for the class-specific accuracy, S3Net gets the highest accuracy in five of nine classes, which also proves the superiority of S3Net. For other methods, DCFSL and SSRN have achieved higher classification accuracy. We consider that this is because these two networks are deeper, which are stronger to extract features for the residual design in SSRN and the transfer learning strategy in DCFSL to get high accuracy.

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TABLE IX

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE INDIAN PINES DATASETS (20 LABELED SAMPLES PER CLASS)

Class	2-D CNN [20]	3-D CNN [46]	SSUN [47]	SSRN [48]	3DAES [36]	DCFSL [39]	D-SSN	S3Net (Ours)
Alfalfa	100.00±0.00	98.55±2.37	100.00±0.00	89.36±9.96	97.83±2.92	100.00±0.00	100.00±0.00	**100.00±0.00**
Corn-notill	60.38±12.78	65.49±6.07	65.86±6.55	87.66±6.56	62.76±8.02	74.09±4.26	85.92±5.69	**88.96±5.87**
Corn-min	68.85±16.18	79.63±5.68	81.12±8.53	84.47±7.63	69.83±8.30	81.23±5.62	88.21±5.24	**89.89±2.87**
Corn	89.08±8.44	93.86±4.44	96.73±2.72	79.39±11.93	87.20±13.43	96.31±3.09	98.11±1.93	**99.49±1.08**
Grass/Pasture	92.20±4.22	89.33±3.80	90.43±3.90	**95.05±2.54**	87.70±6.12	91.84±4.02	92.74±3.97	92.44±4.05
Grass/Trees	96.62±3.55	93.87±4.18	90.15±5.21	96.94±2.18	92.94±3.03	**97.00±1.84**	96.65±3.43	96.63±2.69
Grass/pasture-mowed	100.00±0.00	99.55±1.73	98.57±2.86	63.68±22.03	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
Hay-windrowed	93.30±17.13	99.42±1.54	98.62±2.06	99.60±0.76	98.77±1.60	98.34±1.44	100.00±0.00	**100.00±0.00**
Oats	100.00±0.00	98.75±4.80	100.00±0.00	43.56±19.22	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
Soybeans-notill	74.44±4.79	78.10±4.53	73.16±6.31	76.59±11.34	72.23±4.95	80.15±4.53	87.79±3.94	**88.58±4.82**
Soybeans-min	66.83±6.27	63.34±5.87	71.43±3.43	89.97±3.77	62.15±8.94	72.31±5.33	81.36±5.85	**85.03±5.00**
Soybeans-clean	63.11±4.78	71.76±5.76	80.80±9.70	73.46±10.93	72.06±6.15	78.05±6.27	**94.38±2.59**	92.93±5.08
Wheat	99.57±0.99	98.20±3.02	99.08±1.35	90.60±6.11	**100.00±0.00**	99.68±0.65	99.30±1.11	99.14±1.61
Woods	93.37±3.61	83.07±5.41	88.29±3.67	96.97±2.56	91.64±3.35	93.80±1.61	97.95±2.23	**98.02±3.23**
Bldg-grass-tree-drives	89.15±7.19	90.62±5.82	96.31±2.30	83.59±9.03	82.44±8.16	94.29±4.91	95.22±4.57	**99.07±1.05**
Stone-steel towers	99.86±0.41	98.82±1.52	99.86±0.41	82.68±8.80	**100.00±0.00**	99.59±0.88	99.18±1.10	99.04±1.23
OA (%)	76.77±3.30	77.13±2.29	80.05±1.51	87.09±2.81	75.52±2.60	83.10±1.80	89.91±1.67	**91.54±1.13**
AA (%)	86.67±2.18	87.65±1.05	89.40±0.81	83.35±1.78	86.10±1.93	91.04±0.93	94.80±0.83	**95.58±0.50**
Kappa×100	73.81±3.61	74.25±2.41	77.44±1.66	85.35±3.15	72.32±2.88	80.87±2.00	88.54±1.88	**90.38±1.27**

TABLE X

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE UNIVERSITY OF PAVIA DATASETS (20 LABELED SAMPLES PER CLASS)

Class	2-D CNN [20]	3-D CNN [46]	SSUN [47]	SSRN [48]	3DAES [36]	DCFSL [39]	D-SSN	S3Net (Ours)
Asphalt	80.33±7.60	83.28±5.46	84.70±3.29	91.79±4.27	87.48±3.93	89.74±4.28	94.67±3.40	**95.27±2.59**
Meadows	89.81±4.66	85.96±85.05	81.68±6.30	84.51±8.74	89.51±3.53	91.09±4.42	92.48±6.46	**95.77±4.03**
Gravel	**90.48±5.67**	81.89±5.14	73.12±6.27	85.23±10.94	76.47±6.86	82.93±4.75	89.85±6.23	90.40±6.35
Trees	93.32±3.06	90.57±2.99	**97.72±1.40**	94.1±4.24	97.56±1.01	96.18±2.36	94.16±4.19	96.05±3.20
Painted metal sheets	**100.00±0.00**	99.92±0.12	99.18±1.20	99.99±0.02	99.92±0.14	99.83±0.15	99.98±0.05	99.86±0.13
Bare soil	93.44±3.11	84.93±4.53	83.37±8.29	97.67±1.71	86.16±7.67	93.98±4.03	95.80±3.84	**99.21±1.35**
Bitumen	98.05±2.84	94.90±2.64	89.76±3.64	97.72±2.29	91.04±3.27	90.86±4.16	99.21±0.79	**99.92±0.25**
Self-Blocking Bricks	76.24±7.81	64.40±10.21	92.56±2.99	91.77±9.07	82.44±9.27	85.14±5.09	92.09±5.49	**96.90±3.77**
Shadows	98.33±0.96	60.06±13.26	99.61±0.32	99.76±0.48	99.71±0.56	**99.88±0.10**	99.77±0.24	98.83±0.88
OA (%)	88.64±2.67	83.84±2.14	85.20±2.22	89.75±3.37	88.72±1.82	91.13±1.47	93.77±2.69	**96.27±1.98**
AA (%)	91.11±1.44	82.88±2.00	89.08±1.25	93.62±1.24	90.03±2.06	92.18±1.14	95.34±1.23	**96.91±0.90**
Kappa×100	85.26±3.34	79.02±2.60	80.92±2.63	86.84±4.11	85.29±2.35	88.41±1.85	91.87±3.37	**95.11±2.55**

As shown in Fig. 17, S3Net produces more smooth classification map with higher homogeneity within class and better separability between class. In particular, S3Net obtains the highest accuracy when classifying meadows and bare soil, whereas the other methods have many missclassified pixels, which is in accordance with the results reported in Table X. In the unlabeled areas on the right side of the data, S3Net can also distinguish different samples more correctly. Another observation is that it is more in line with the distribution of real objects.

According to the result of Salinas dataset reported in Table XI, S3Net and D-SSN outperform the other methods with the highest accuracies. Specifically, S3Net obtains the highest OA of 95.95%, which achieves significant improvements ranging from 1.51% to 3.59%. In addition, S3Net also produces significant improvements in terms of AA and Kappa. As for the class-specific accuracy, S3Net has higher accuracy in 7 of 16 classes. It is interesting to note that other methods have also achieved high classification accuracies with little difference, which is because of the regular ground object distribution in this scene.

As shown in Fig. 18, S3Net gets better classification map with higher homogeneity within class and separability between class. Especially in Fig. 18(i), S3Net obtains the smoothest results for vineyard untrained, whereas the other methods produce worse results. In the unlabeled area on the lower right side of the data, S3Net classifies these samples more accurately than other methods.

Although larger input blocks are used in S3Net, it does not cause large test error evaluations from the results. We consider that one branch of S3Net uses a relatively small convolution kernel of 7×7 and adaptive weight fusion is used when the two branches are fused; thus, the network can be trained during training to learn the best fusion ratio. In this context, the test evaluation errors of the model in the unlabeled part are reduced and the method can get smoother full classification maps. Therefore, we consider that S3Net can alleviate the spatial autocorrelation issue.

As we can see, our method outperforms the other methods with the highest accuracies in three datasets. In S3Net, we use a lightweight network to construct a Siamese network framework and design a differentiated input strategy to provide more features for the model. In addition, the weighted contrastive loss and adaptive cross-entropy loss can further enhance the convergence ability and expression ability of the model, and thereby, our method achieves the highest classification accuracies. An interesting observation is that SSRN obtains higher averaged OA in Indian Pines, whereas the network gets the

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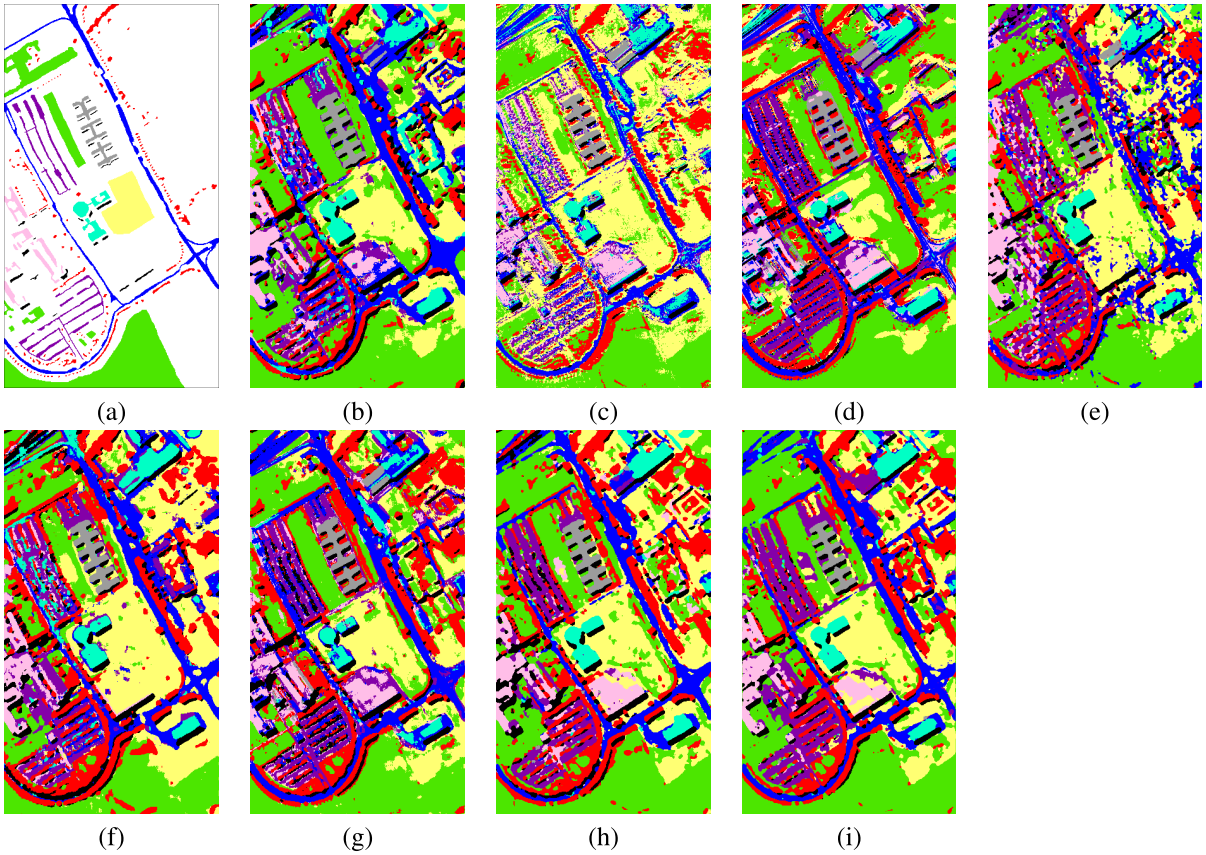


Fig. 17. Classification maps obtained by different methods for the University of Pavia dataset (20 labeled samples per class). (a) Ground truth. (b) 2-D CNN (88.64%). (c) 3-D CNN (83.84%). (d) SSUN (85.20%). (e) SSRN (89.75%). (f) 3DAES (88.72%). (g) DCFSL (91.13%). (h) D-SSN (93.77%). (i) S3Net (96.27%).

TABLE XI
CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR THE SALINAS DATASETS (20 LABELED SAMPLES PER CLASS)

Class	2-D CNN [20]	3-D CNN [46]	SSUN [47]	SSRN [48]	3DAES [36]	DCFSL [39]	D-SSN	S3Net (Ours)
Broccoli_green_weeds_1	98.90±1.87	99.41±0.45	97.05±3.27	**100.00±0.00**	99.86±0.23	99.69±0.45	99.96±0.12	99.97±0.08
Broccoli_green_weeds_22	99.98±0.06	99.46±0.60	95.95±3.86	99.64±0.56	99.40±1.41	99.92±0.08	99.94±0.13	**99.99±0.02**
Fallow	99.74±0.33	99.61±0.39	98.61±1.23	99.89±0.16	98.26±5.22	99.86±0.24	100.00±0.00	**100.00±0.00**
Fallow_rough_plow	98.72±1.00	99.17±0.60	**99.88±0.20**	97.81±1.90	99.80±0.40	99.50±0.34	99.80±0.23	99.60±0.96
Fallow_smooth	96.12±2.22	98.00±1.85	98.54±0.47	98.17±1.84	96.15±4.00	96.00±2.90	**98.96±0.93**	98.54±1.40
Stubble	99.76±0.49	99.83±0.51	99.72±0.32	99.94±0.14	99.24±1.08	99.90±0.16	**99.99±0.01**	99.98±0.03
Celery	99.40±1.03	99.20±0.60	97.13±2.35	99.81±0.49	99.95±0.07	99.47±0.38	**99.99±0.03**	99.98±0.06
Grapes_untrained	81.90±5.76	78.79±4.51	80.54±4.56	88.15±3.10	78.15±9.40	85.27±3.47	85.31±6.20	**88.07±8.00**
Soil_vineyard_develop	99.85±0.27	99.40±0.40	99.56±0.35	99.56±0.23	99.95±0.07	99.89±0.07	100.00±0.00	**100.00±0.00**
Corn_senesced_green_weeds	95.52±2.41	92.89±2.86	95.60±2.17	96.34±2.10	94.02±3.07	95.37±1.56	**97.08±2.12**	96.58±1.87
Lettuce_romaine_4wk	99.81±0.24	98.85±0.61	98.38±0.89	97.30±3.32	99.94±0.11	99.32±0.72	99.86±0.26	**99.90±0.14**
Lettuce_romaine_5wk	99.62±0.68	99.75±0.56	99.77±0.56	99.04±1.15	99.09±1.53	**99.97±0.06**	99.92±0.17	99.57±0.68
Lettuce_romaine_6wk	99.34±0.67	99.64±0.46	99.38±0.51	98.72±1.60	97.12±3.60	99.48±0.38	99.12±1.81	**99.81±0.32**
Lettuce_romaine_7wk	**99.36±0.47**	**99.36±0.40**	98.60±1.30	95.09±5.83	97.81±3.13	99.34±0.65	97.66±4.03	99.07±1.47
Vineyard_untrained	80.50±6.79	87.82±3.04	84.39±8.91	79.10±4.78	83.72±12.53	81.51±4.47	84.13±9.72	**91.00±6.18**
Vineyard_vertical_trellis	98.35±1.27	96.85±4.19	97.08±2.29	98.93±2.62	98.68±0.85	98.44±1.18	99.08±0.61	**99.54±0.50**
OA (%)	92.87±0.77	93.01±0.90	92.61±0.89	93.83±0.71	92.36±1.06	93.78±0.54	94.45±0.59	**95.95±1.27**
AA (%)	96.68±0.42	96.75±0.39	96.26±0.78	96.72±0.80	96.20±0.61	97.06±0.30	97.54±0.44	**98.23±0.41**
Kappa×100	92.08±0.85	92.24±0.99	91.79±1.00	93.13±0.79	91.51±1.17	93.07±0.60	93.83±0.66	**95.49±1.40**

highest standard deviation for most classes. This observation illustrates that even SSRN has strong feature extraction ability by using a deeper residual network to get high classification accuracy, but the network has an unstable training process due to the large parameter size, which causes an overfitting issue.

H. Complexity Analysis

To analyze the complexity of S3Net, we compare 2-D CNN, 3-D CNN, SSUN, SSRN, 3DAES, DCFSL, and D-SSN using five labeled samples per class in three datasets. The obtained training time, test time, and parameter size are reported in Table XII. As we can see, D-SSN has the fewest parameters with relatively short training and test time compared to the other DL methods. This is because the subnetwork of D-SSN uses 1-D and 2-D convolution to replace 3-D convolution and uses GAP to replace the fully connected layer, which effectively reduces network parameter size. S3Net has more parameters than D-SSN, which is because one-layer weights

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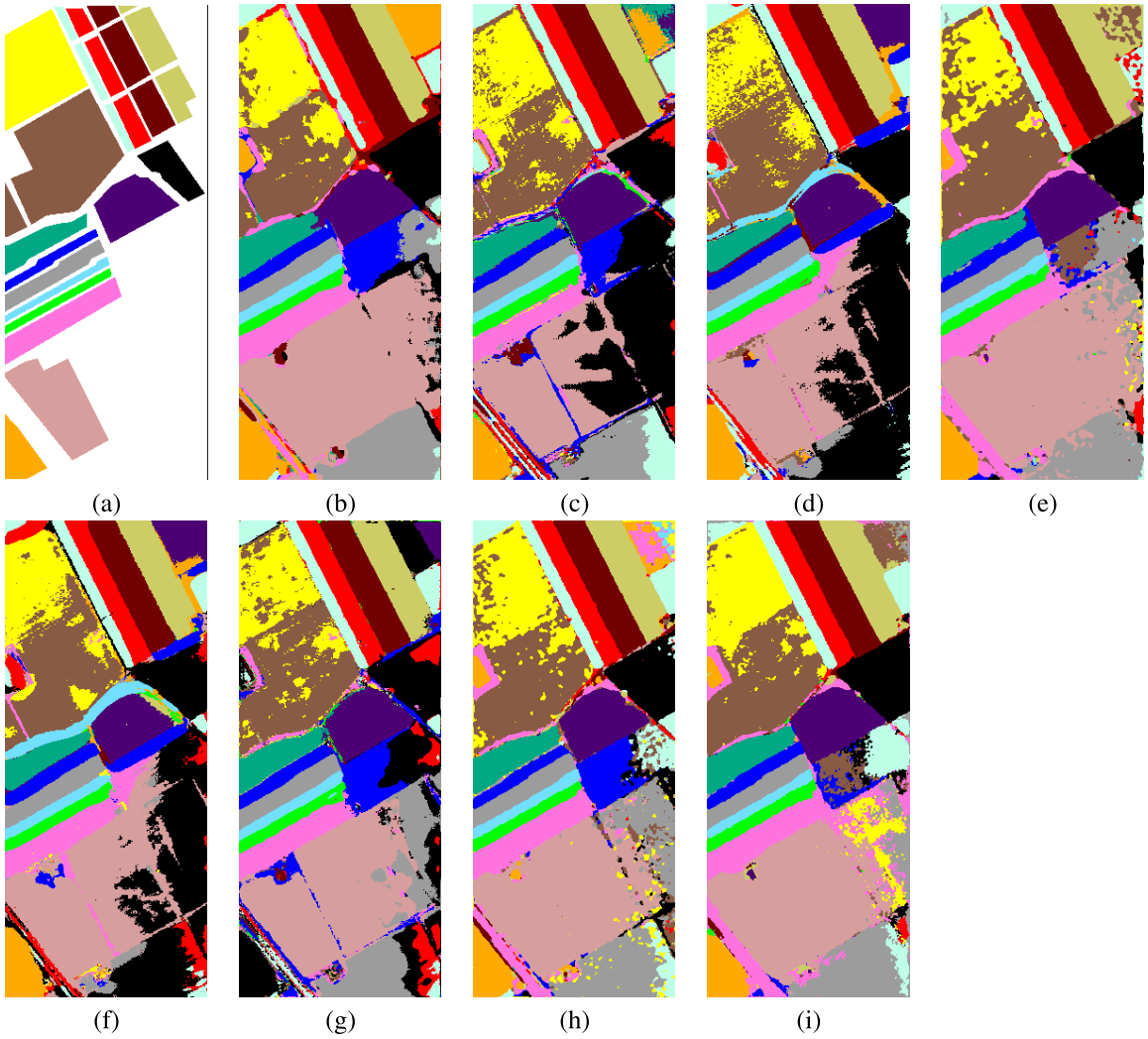


Fig. 18. Classification maps obtained by different methods for the Salinas dataset (20 labeled samples per class). (a) Ground truth. (b) 2-D CNN (92.87%). (c) 3-D CNN (93.01%). (d) SSUN (92.61%). (e) SSRN (93.83%). (f) 3DAES (92.36%). (g) DCFSL (93.78%). (h) D-SSN (94.45%). (i) S3Net (95.95%).

in SSN of two branches are unshared, but the parameter size of S3Net is still small than the other methods, which is also suitable for small sample classification.

V. Discussion

In this section, we discuss the separability of feature maps obtained by different methods, including SSN, D-SSN, and S3Net. In addition, we give some considerations on overfitting issue under small training samples.

A. Feature Separability

To display the separability of different feature maps, we use PCA to transform the output features into the 2-D space, as shown in Fig. 19. According to the result, for University of Pavia and Indian Pines datasets, it can be clearly seen that the feature separability of the D-SSN is better than SSN, and the feature separability of S3Net is the best. It is not particularly obvious in Salinas because the distribution of different land covers in Salinas data is more regular, and thus, different methods can accurately distinguish different classes. According to the 2-D scatter plot, it can be demonstrated that D-SSN can reduce the distance within class and increase the distance between class by using contrastive loss in the Siamese network framework. Compared with D-SSN, S3Net with other proposed ingredients further improves the separability of feature maps, thus achieving the highest classification accuracy.

In Fig. 20, the first column is a false-color composite image, and the other columns are the feature maps of SSN, D-SSN, and S3Net. The feature map is composed of the last layer feature for each model and the original image with a composition ratio of 3:7. According to the figure, the features extracted by SSN are generally messy, and they cannot accurately reflect the content of the input patch. Siamese network realizes sample amplification through pairing, and it controls the distance between samples through the contrastive loss, thus improving the classification ability of the model. After using Siamese network, the feature map can generally describe the content of input patch. Compared with D-SSN, S3Net uses a differentiated input strategy to provide more features for the model, whereas the weighted contrastive loss

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TABLE XII

RUNNING TIME (S) AND PARAMETER SIZE OF DIFFERENT METHODS FOR THE THREE DATASETS (FIVE LABELED SAMPLES PER CLASS)

Data sets	Metrics	2-D CNN [20]	3-D CNN [46]	SSUN [47]	SSRN [48]	3DAES [36]	DCFSL [39]	D-SSN	S3Net (Ours)
Indian Pines	Training time (s)	3.10	36.51	29.74	13.91	2.31	3500	3.89	3.99
	Test time (s)	0.42	1.10	0.89	11.92	1.43	10	1.82	2.12
	Parameter size	115152	1539024	1215888	363768	269584	4259294	102832	164464
University of Pavia	Training time (s)	4.14	40.03	43.43	5.1	1.14	2300	2.11	2.27
	Test time (s)	1.54	2.19	2.64	62.38	1.08	56	1.31	1.65
	Parameter size	107081	880073	1196795	216312	138128	4270121	70061	100973
Salinas	Training time (s)	4.43	47.22	49.43	13.3	2.76	2200	3.93	4.13
	Test time (s)	1.94	3.69	3.33	63.60	2.03	40	2.11	3.21
	Parameter size	115152	1539024	1216912	369912	269584	4270121	102832	164464

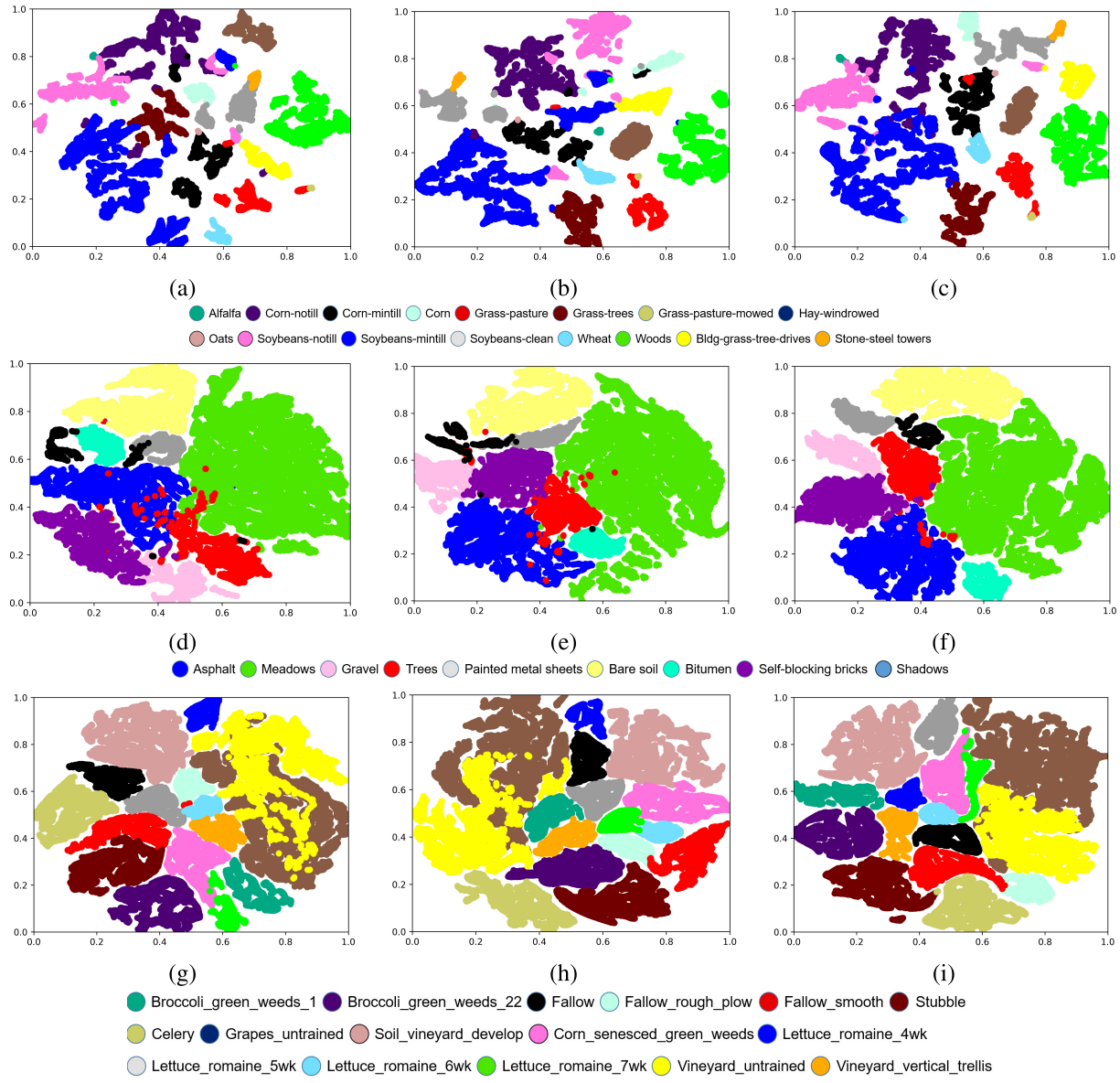


Fig. 19. Feature separability for different methods in the three datasets. (a) SSN, (b) D-SSN, and (c) S3Net for Indian Pines. (d) SSN, (e) D-SSN, and (f) S3Net for University of Pavia. (g) SSN, (h) D-SSN, and (i) S3Net for Salinas.

and adaptive cross-entropy loss improve the feature extraction ability, thus achieving significant performance improvements.

B. Considering Overfitting

In the HSI classification task, the overfitting problem is jointly caused by too many parameters of the model and insufficient training samples. Although some deep networks have powerful feature extraction capability due to deep structures, they lead to high standard deviations under the training condition of small samples. This is because they usually have too many parameters, which results in an overfitting issue. To address the issue of large parameter size, we use 1-D

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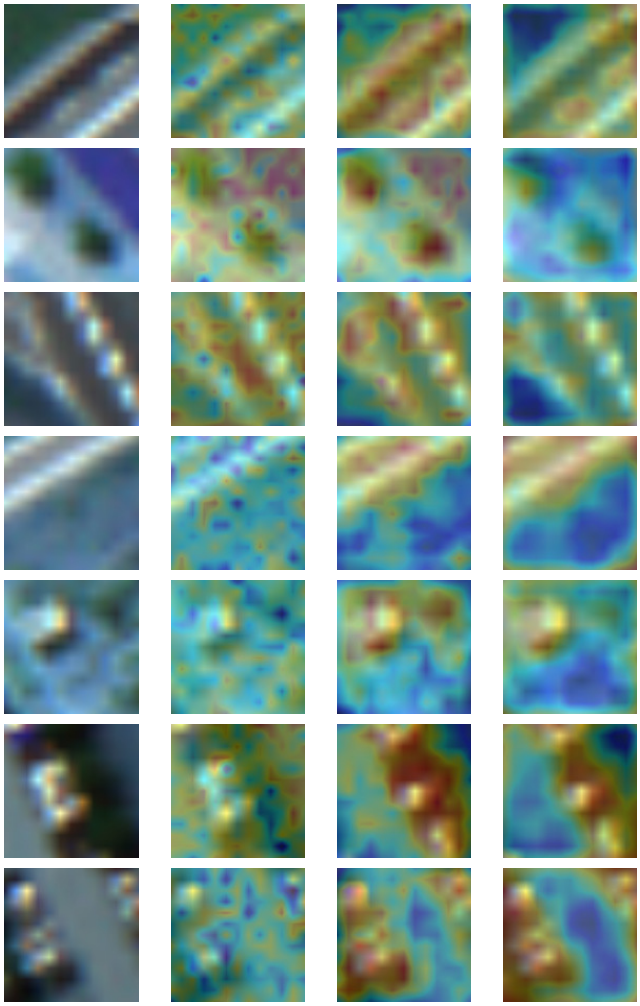


Fig. 20. Feature maps of the proposed methods in University of Pavia. (a) Image patch. (b) SSN. (c) D-SSN. (d) S3Net.

and 2-D convolutions to extract spectral and spatial features, which can reduce the amount of parameters by 2/3 compared with using 3-D convolution. To solve the problem of limited training samples, we use the Siamese network to expand the sample size, and we design a differentiated input strategy, which can improve more features to the model. In addition, the adaptive cross-entropy loss and the weighted contrastive loss can improve the feature extraction ability of the model under small samples. Therefore, the method proposed in this article is not prone to overfitting under small samples.

VI. Conclusion

In this article, we propose a novel S3Net for HSI classification. The innovative contribution of S3Net lies in the design of the lightweight SSN, differentiated input strategy for Siamese network framework, and weighted contrastive loss and adaptive cross-entropy loss, which jointly contribute to enhance the discrimination power of S3Net. Experimental results conducted on three HSI datasets validated that our method effectively weakens the overfitting issue and achieves high classification accuracy compared with traditional and other state-of-the-art DL-based HSI classification methods under different training samples.

Although we have obtained encouraged experimental results, we will continue our research by integrating active learning to obtain more effective negative samples. Currently, we use RS to prepare the sample pairs, which may not be optimal since the quality of negative samples cannot be guaranteed, thus affecting the converging speed and classification performance, whereas probability-based active learning can be easily equipped with our model, which can accelerate the converging speed and may also improve the classification performance.

Last but not least, a very deep network is more prone to face overfitting, but lightweight network may have poor capacity to effectively learn features. Therefore, it is significant to balance between the model complexity and feature learning capability when designing deep architectures. In this context, the relationship between model complexity and feature learning capability for few-shot HSI classification deserves careful exploration.

Acknowledgment

The authors would like to thank Prof. D. Landgrebe for making the Airborne Visible/Infrared Imaging Spectrometer Indian Pines hyperspectral dataset available to the community and Prof. P.Gamba for providing the Reflective Optics Spectrographic Imaging System data over Pavia, Italy.

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